

Topoi (ILF)

Goldblatt / Exercises / Reading group

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Part I

Introduction

Meeting II

Category Theory Pt. 1

1.1 Minimal Exercises

1.1.1 Pre-order Category

For each natural number n , consider the pre-order category $\mathbf{n}$ associated with the pre-order $0 \leq 1 \leq \dots \leq n-1$. How many arrows does $\mathbf{n}$ have?

Solution

For each natural number $k < n$, we can count the number of incoming arrows as

$$|\{j \in \{0, 1, \dots, n-1\} : j \leq k\}| = k + 1.$$

Therefore, there are

$$\sum_{k=0}^{n-1} (k+1) = \binom{n}{2}$$

arrows in total.

1.1.2 Subcategories of a Monoid

Consider the monoid $\mathbf{N} = (N, +, 0)$ of natural numbers, which gives rise to a category whose collection of objects is the singleton $\{N\}$ and whose collection of arrows is the set N of natural numbers. Which of the following $\mathcal{C}$ are valid subcategories?

- The empty category $\mathcal{C} = \mathbf{0}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{0\} = \{1, 2, 3, 4, \dots\}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{1\} = \{0, 2, 3, 4, \dots\}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{2\} = \{0, 1, 3, 4, \dots\}$

Solution

- The empty category $\mathcal{C} = \mathbf{0}$

The two conditions for being a subcategory are vacuously true; in fact, the empty category is a

subcategory of every other category. Let $\mathcal{D}$ be any category. Then:

- (i) Every $\mathbf{0}$ -object is a $\mathcal{D}$ -object, vacuously.
- (ii) For every pair of $\mathbf{0}$ -objects a, b , the inclusion $\mathbf{0}(a, b) \subseteq \mathcal{D}(a, b)$ holds, vacuously.

- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{0\} = \{1, 2, 3, 4, \dots\}$

Subcategories must be closed under identities. Since the identity is omitted from the given subquiver, it is not a subcategory.

- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{1\} = \{0, 2, 3, 4, \dots\}$

The given data already form a subquiver. It suffices to check that the subquiver is a category itself with the inherited identity and composition operations. This amounts to checking that the set $N \setminus \{1\}$ is a submonoid of the natural numbers.

1. Since $0 \in N \setminus \{1\}$, it is closed under identities.
2. Let $x, y \in N \setminus \{1\}$. If either $x = 0$ or $y = 0$, then $x + y \in \{x, y\} \subseteq N \setminus \{1\}$. The remaining case is when neither are 0, in which case $x, y \geq 2$. Therefore, $x + y \geq 4$ and so $x + y \in N \setminus \{1\}$.
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{2\} = \{0, 1, 3, 4, \dots\}$

The given subquiver is not closed under composition: $1 \in N \setminus \{2\}$ yet $1+1 \notin N \setminus \{2\}$. Therefore, the given subquiver is not a subcategory.

1.2 Exercises for Good Preparation

1.2.1 Endomorphism Monoid

Let $\mathcal{C}$ be a category and X be an object of $\mathcal{C}$. Verify that the collection $\mathcal{C}(X, X)$ of all $\mathcal{C}$ -arrows from X to X forms a monoid with respect to composition.

Solution

We shall check that $(\mathcal{C}(X, X), \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)}, 1_X)$ is a monoid.

Notes

It is worth mentioning that the “hom-collection” $\mathcal{C}(X, X)$ need not be a “small” set whenever $\mathcal{C}$ is not a “locally small” category. In that case, we would call $\mathcal{C}(X, X)$ a “large” (i.e. proper class-sized) monoid.

Let $\mathcal{C}_1$ denote the collection of arrows of $\mathcal{C}$. Notice that $\mathcal{C}(X, X) \times \mathcal{C}(X, X) \subseteq \mathcal{C}_1^{\text{cod} \times \text{dom}} \mathcal{C}_1$

and that $\circ(\mathcal{C}(X, X) \times \mathcal{C}(X, X)) \subseteq \mathcal{C}(X, X)$. Therefore, $\circ$ restricts to a binary operation on the collection $\mathcal{C}(X, X)$.

Notes

It is also worth mentioning that Goldblatt defines composition as an operation with typing $\circ : \mathcal{C}_1^{\text{dom} \times \text{cod}} \mathcal{C}_1 \rightarrow \mathcal{C}_1$, whereas we are using the typing $\circ : \mathcal{C}_1^{\text{cod} \times \text{dom}} \mathcal{C}_1 \rightarrow \mathcal{C}_1$. The two operations are essentially the same, up to swapping arguments.

The associativity of composition in a category translates to associativity of composition on subset $\mathcal{C}(X, X)$. Pedantically, for $f, g, h \in \mathcal{C}(X, X)$:

$$\begin{aligned} & h \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} (g \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} f) \\ &= h \circ (g \circ f) \\ &= (h \circ g) \circ f \\ &= (h \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} g) \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} f \end{aligned}$$

Likewise, $1_X \in \mathcal{C}(X, X)$ acts as an identity for compositions from $\mathcal{C}_1$ that are defined. Therefore, 1_X also acts as an identity on the subset $\mathcal{C}(X, X)$, for which composition is always defined. Pedantically, for $f \in \mathcal{C}(X, X)$:

$$\begin{aligned} 1_X \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} f &= 1_X \circ f = f \\ f \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} 1_X &= f \circ 1_X = f \end{aligned}$$

Therefore, $(\mathcal{C}(X, X), \circ, 1_X)$ is a monoid (modulo the fact that the collection $\mathcal{C}(X, X)$ can be a proper class).

1.2.2 Sum Category

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. The sum category, denoted $\mathcal{C} + \mathcal{D}$, is a category whose collection of objects is the disjoint union of $\mathcal{C}$ -objects and $\mathcal{D}$ -objects, and whose collection of arrows is the disjoint union of $\mathcal{C}$ -arrows and $\mathcal{D}$ -arrows.

- Describe how to inherit domains, codomains, identity, and composition from both $\mathcal{C}$ and $\mathcal{D}$. Verify that $\mathcal{C} + \mathcal{D}$ is a category.
- Verify that there are no $(\mathcal{C} + \mathcal{D})$ -arrows between any object that came from $\mathcal{C}$ and any object that came from $\mathcal{D}$.
- Find an example of $\mathcal{C}$ and $\mathcal{D}$ where if set union was used instead of disjoint union, then the resulting construction would not yield a category.

Notes

For collections A and B , let $A + B$ denote the disjoint union. To be pedantic:

$$A + B = \{\langle a, 0 \rangle : a \in A\} \cup \{\langle b, 1 \rangle : b \in B\}$$

If either A or B is a “large” set (i.e. a proper class), then $A + B$ is also a “large” set. Also note that $A + B$ is “small” whenever A and B are both “small”.

For operations $f : A \rightarrow C$ and $g : B \rightarrow D$ between collections, let $f + g : A + B \rightarrow C + D$ denote the operation defined as follows:

$$(f + g)(\langle a, 0 \rangle) = \langle f(a), 0 \rangle$$

$$(f + g)(\langle b, 1 \rangle) = \langle f(b), 1 \rangle$$

Solution Part I

Let $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$ and $\mathcal{D} = (O', A', \text{dom}', \text{cod}', \circ', 1')$ be the two categories. The typing of our operations is as follows:

$$\text{dom} : A \rightarrow O$$

$$\text{dom}' : A' \rightarrow O'$$

$$\text{cod} : A \rightarrow O$$

$$\text{cod}' : A' \rightarrow O'$$

$$\circ : A_{\text{dom} \times \text{cod}} A \rightarrow A$$

$$\circ' : A'_{\text{dom}' \times \text{cod}'} A' \rightarrow A'$$

$$1 : O \rightarrow A$$

$$1' : O' \rightarrow A'$$

(Here, we choose the typing of $\circ$ and $\circ'$ to follow Goldblatt’s convention: $\circ(\langle g, f \rangle) = g \circ f$ whenever $f : a \rightarrow b$ and $g : b \rightarrow c$.)

As a first approximation, we wish to claim that $\mathcal{C} + \mathcal{D} = (O + O', A + A', \text{dom} + \text{dom}', \text{cod} + \text{cod}', \circ + \circ', 1 + 1')$ is a valid category. The proposed operations would have the following typing:

$$\text{dom} + \text{dom}' : A + A' \rightarrow O + O'$$

$$\text{cod} + \text{cod}' : A + A' \rightarrow O + O'$$

$$\circ + \circ' : (A_{\text{dom} \times \text{cod}} A) + (A'_{\text{dom}' \times \text{cod}'} A') \rightarrow A + A'$$

$$1 + 1' : O + O' \rightarrow A + A'$$

This is close to what we need, but our proposed composition does not have the required typing. To be pedantic, we can adjust the domain of $\circ + \circ'$ by precomposing with the following (bijective) operation:

$$(A + A')_{\text{dom} + \text{dom}' \times \text{cod} + \text{cod}'} (A + A') \xrightarrow{\phi} (A_{\text{dom} \times \text{cod}} A) + (A'_{\text{dom}' \times \text{cod}'} A')$$

$$\langle \langle g, 0 \rangle, \langle f, 0 \rangle \rangle \longmapsto \langle \langle g, f \rangle, 0 \rangle \quad \text{for } a \xrightarrow{f} b \xrightarrow{g} c \text{ in } \mathcal{C}$$

$$\langle \langle g', 1 \rangle, \langle f', 1 \rangle \rangle \longmapsto \langle \langle g', f' \rangle, 1 \rangle \quad \text{for } a' \xrightarrow{f'} b' \xrightarrow{g'} c' \text{ in } \mathcal{D}$$

To verify that we have specified all possible input combinations in our definition of ϕ , note that for each $h \in A + A'$ and $n \in \{0, 1\}$, we have that $(\text{dom} + \text{dom}')(\langle h, n \rangle) = n = (\text{cod} + \text{cod}')(\langle h, n \rangle)$.

Therefore:

$$\begin{aligned}
 & (A + A')_{\text{dom} + \text{dom}' \times \text{cod} + \text{cod}'} (A + A') \\
 &= \{ \langle \langle h, m \rangle, \langle k, n \rangle \rangle \in (A + A')^2 : (\text{dom} + \text{dom}')(\langle h, m \rangle) = (\text{cod} + \text{cod}')(\langle k, n \rangle) \} \\
 &= \{ \langle \langle h, m \rangle, \langle k, n \rangle \rangle \in (A + A')^2 : m = n \}
 \end{aligned}$$

Now, we claim that $\mathcal{C} + \mathcal{D} = (O + O', A + A', \text{dom} + \text{dom}', \text{cod} + \text{cod}', (\circ + \circ')\phi, 1 + 1')$ satisfies the axioms of a category. We do so as follows:

- Domain of Composition:

$$\begin{aligned}
 & (\text{dom} + \text{dom}')((\circ + \circ')\phi(\langle \langle g, 0 \rangle, \langle f, 0 \rangle \rangle)) \\
 &= (\text{dom} + \text{dom}')((\circ + \circ')(\langle \langle g, f \rangle, 0 \rangle)) \\
 &= (\text{dom} + \text{dom}')(\langle \circ(\langle g, f \rangle), 0 \rangle) \\
 &= \langle \text{dom}(\circ(\langle g, f \rangle)), 0 \rangle \\
 &= \langle \text{dom}(g \circ f), 0 \rangle \\
 &= \langle \text{dom}(f), 0 \rangle \\
 &= (\text{dom} + \text{dom}')(\langle f, 0 \rangle)
 \end{aligned}$$

and by symmetry (with computation omitted):

$$(\text{dom} + \text{dom}')((\circ + \circ')\phi(\langle \langle g', 1 \rangle, \langle f', 1 \rangle \rangle)) = (\text{dom} + \text{dom}')(\langle f', 1 \rangle)$$

- Codomain of Composition:

Computations follow by a symmetric argument to verifying domain of composition. With computations omitted, we have the following:

$$(\text{cod} + \text{cod}')((\circ + \circ')\phi(\langle \langle g, 0 \rangle, \langle f, 0 \rangle \rangle)) = (\text{cod} + \text{cod}')(\langle g, 0 \rangle)$$

and

$$(\text{cod} + \text{cod}')((\circ + \circ')\phi(\langle \langle g', 1 \rangle, \langle f', 1 \rangle \rangle)) = (\text{cod} + \text{cod}')(\langle g', 1 \rangle)$$

- Associativity:

$$\begin{aligned}
 & (\circ + \circ')\phi((\circ + \circ')\phi(\langle \langle h, 0 \rangle, \langle g, 0 \rangle \rangle), \langle f, 0 \rangle) \\
 &= (\circ + \circ')\phi((\circ + \circ')(\langle \langle h, g \rangle, 0 \rangle), \langle f, 0 \rangle) \\
 &= (\circ + \circ')\phi(\langle \circ(\langle h, g \rangle), 0 \rangle, \langle f, 0 \rangle) \\
 &= (\circ + \circ')(\langle \circ(\langle h, g \rangle), f \rangle, 0) \\
 &= \langle \circ(\circ(\langle h, g \rangle), f), 0 \rangle \\
 &= \langle \circ(\langle h \circ g, f \rangle), 0 \rangle \\
 &= \langle (h \circ g) \circ f, 0 \rangle
 \end{aligned}$$

and by symmetry (with computation omitted):

$$(\circ + \circ')\phi(\langle \langle h, 0 \rangle, (\circ + \circ')\phi(\langle \langle g, 0 \rangle, \langle f, 0 \rangle \rangle) \rangle) = \langle h \circ (g \circ f), 0 \rangle$$

Since $(h \circ g) \circ f = h \circ (g \circ f)$ holds in $\mathcal{C}$, it follows that

$$\begin{aligned} & (\circ + \circ')\phi(\langle(\circ + \circ')\phi(\langle\langle h, 0 \rangle, \langle g, 0 \rangle\rangle), \langle f, 0 \rangle\rangle) \\ &= \langle(h \circ g) \circ f, 0\rangle \\ &= \langle h \circ (g \circ f), 0\rangle \\ &= (\circ + \circ')\phi(\langle\langle h, 0 \rangle, (\circ + \circ')\phi(\langle\langle g, 0 \rangle, \langle f, 0 \rangle\rangle)\rangle) \end{aligned}$$

and symmetrically (with computation omitted) for

$$\begin{aligned} & (\circ + \circ')\phi(\langle(\circ + \circ')\phi(\langle\langle h', 1 \rangle, \langle g', 1 \rangle\rangle), \langle f', 1 \rangle\rangle) \\ &= \langle(h' \circ g') \circ f', 1\rangle \\ &= \langle h' \circ (g' \circ f'), 1\rangle \\ &= (\circ + \circ')\phi(\langle\langle h', 1 \rangle, (\circ + \circ')\phi(\langle\langle g', 1 \rangle, \langle f', 1 \rangle\rangle)\rangle) \end{aligned}$$

- Domain of identity:

$$\begin{aligned} & (\text{dom} + \text{dom}')((1 + 1')(\langle a, 0 \rangle)) & (\text{dom} + \text{dom}')((1 + 1')(\langle a', 1 \rangle)) \\ &= (\text{dom} + \text{dom}')(\langle 1(a), 0 \rangle) &= (\text{dom} + \text{dom}')(\langle 1'(a'), 1 \rangle) \\ &= \langle \text{dom}(1(a)), 0 \rangle &= \langle \text{dom}'(1'(a')), 1 \rangle \\ &= \langle a, 0 \rangle &= \langle a', 1 \rangle \end{aligned}$$

- Codomain of identity:

$$\begin{aligned} & (\text{cod} + \text{cod}')((1 + 1')(\langle a, 0 \rangle)) & (\text{cod} + \text{cod}')((1 + 1')(\langle a', 1 \rangle)) \\ &= (\text{cod} + \text{cod}')(\langle 1(a), 0 \rangle) &= (\text{cod} + \text{cod}')(\langle 1'(a'), 1 \rangle) \\ &= \langle \text{cod}(1(a)), 0 \rangle &= \langle \text{cod}'(1'(a')), 1 \rangle \\ &= \langle a, 0 \rangle &= \langle a', 1 \rangle \end{aligned}$$

- Right identity:

$$\begin{aligned} & (\circ + \circ')\phi(\langle\langle f, 0 \rangle, (1 + 1')((\text{dom} + \text{dom}')(\langle f, 0 \rangle))\rangle) \\ &= (\circ + \circ')\phi(\langle\langle f, 0 \rangle, (1 + 1')(\langle \text{dom}(f), 0 \rangle)\rangle) \\ &= (\circ + \circ')\phi(\langle\langle f, 0 \rangle, \langle 1(\text{dom}(f)), 0 \rangle\rangle) \\ &= (\circ + \circ')(\langle\langle f, 1(\text{dom}(f)) \rangle, 0 \rangle) \\ &= \langle \circ(\langle f, 1(\text{dom}(f)) \rangle), 0 \rangle \\ &= \langle f, 0 \rangle \end{aligned}$$

and symmetrically (with computation omitted) for

$$(\circ + \circ')\phi(\langle\langle f', 1 \rangle, (1 + 1')((\text{dom} + \text{dom}')(\langle f', 1 \rangle))\rangle) = \langle f', 1 \rangle$$

- Left identity:

Computations follow by a symmetric argument to verifying right identity. With computations omitted, we have the following:

$$(\circ + \circ')\phi(\langle(1 + 1')((\text{dom} + \text{dom}')(\langle f, 0 \rangle)), \langle f, 0 \rangle\rangle) = \langle f, 0 \rangle$$

and

$$(\circ + \circ')\phi(\langle(1 + 1')((\text{dom} + \text{dom}')(\langle f', 1 \rangle)), \langle f', 1 \rangle\rangle) = \langle f', 1 \rangle$$

Solution Part II

This follows from the construction above. If $\langle h, m \rangle$ is composable with $\langle k, n \rangle$, then $m = n$ follows, as shown above.

Solution Part III

Consider the categories $(\{0\}, \{1\}, !, !, !, !)$ and $(\{0\}, \{2\}, !, !, !, !)$ where $!$ is a placeholder for the unique possible function in that slot. Had we used set union instead of disjoint union, we would have obtained a quiver with one vertex 0 and two edges 1 and 2. Although we would have $1 \circ 1 = 1$ and $2 \circ 2 = 2$, it is unclear how to define $1 \circ 2$ and $2 \circ 1$. In particular, the ϕ used in the construction above would no longer be defined on all inputs.

1.3 Extra Exercise

1.3.1 Quivers

A *quiver* is a directed graph that allows loops and multiple parallel edges. In particular, a quiver Q comprises:

- a collection of things called Q -vertices;
- a collection of things called Q -edges;
- operations assigning to each Q -edge f a Q -vertex $s(f)$ (the “source” of f) and a Q -vertex $t(f)$ (the “target” of f).

Every category has an underlying quiver by forgetting the composition (and identity): each object is a vertex, each arrow is an edge, $s(f) = \text{dom}(f)$, and $t(f) = \text{cod}(f)$.

Two quivers are considered isomorphic when there is a bijection of vertices and a bijection of edges that preserves sources and targets. Two categories are considered isomorphic when there exists an isomorphism of the underlying quivers that additionally preserves composition and identity.

Find an example of two non-isomorphic categories such that the underlying quivers are isomorphic.

Solution

There are many possible solutions to this exercise. Here, we'll cover the smallest solution:

Let $M = (\mathbb{Z}/2\mathbb{Z}, +, 0)$ and $N = (\mathbb{Z}/2\mathbb{Z}, \cdot, 1)$ be the additive monoid and multiplicative monoid of $\mathbb{Z}/2\mathbb{Z}$, respectively. The categories associated to these monoids each have one object and two arrows. Therefore, their underlying quivers are isomorphic.

The second category satisfies the first-order formula $(\forall f)(f \circ f = f)$ as f varies over all arrows, whereas the first category does not. Therefore, the two categories cannot be isomorphic.